Supervised Learning for Medellín Land Cover Classification

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Overview

Phenomenon:

Corredores Verdes program (since 2016)

for Urban heat island mitigation in Medellin

Remote Sensing & Machine Learning

Land cover classification:

- Random Forest
- Support vector Machine

Best model test on Bogotá AOI

Change detection:

- Medellin AOI
- Green corridors
- Thermal analysis

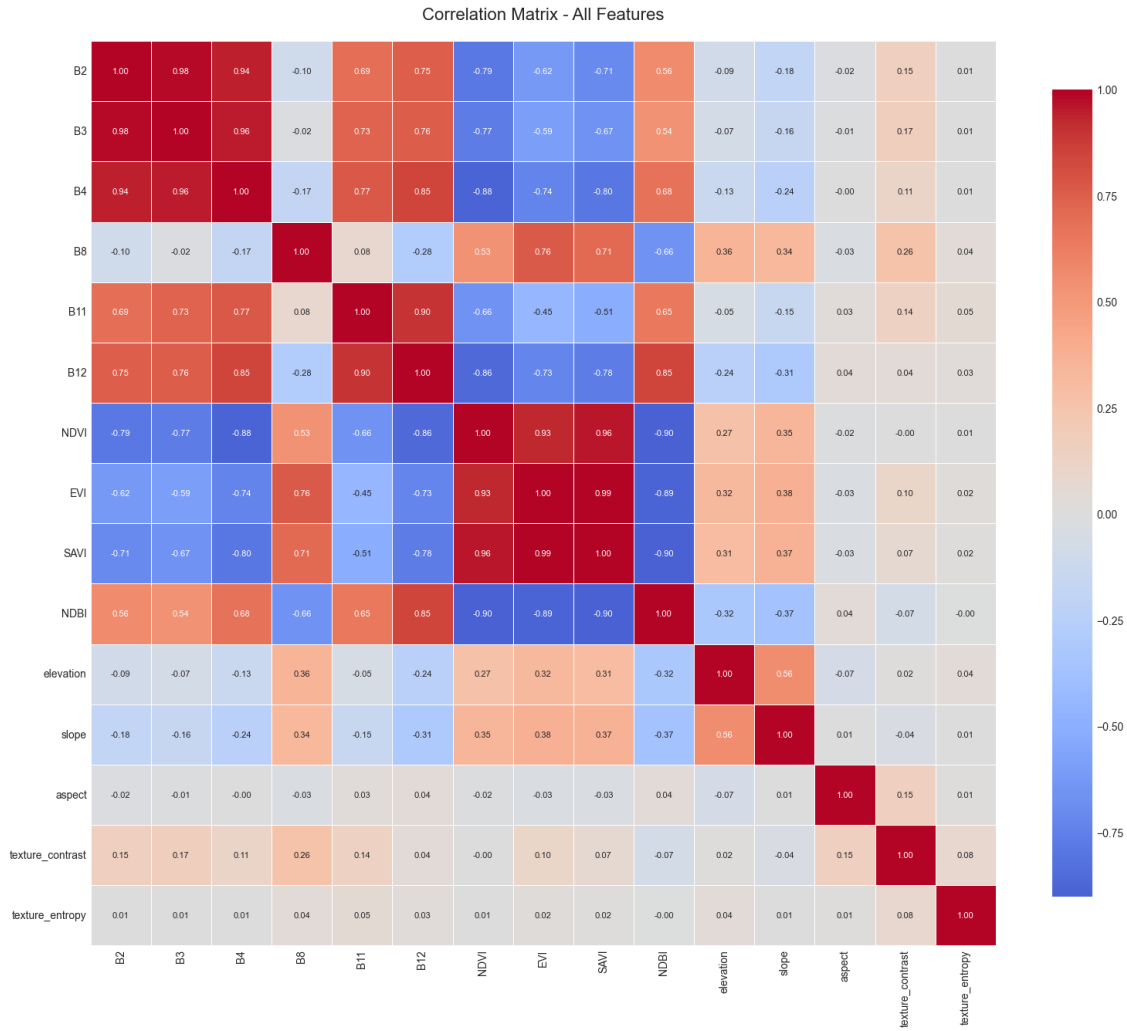
Materials and Data

Images and Source	Sentinel 2 for Land cover and Landsat 8 for Thermal analysis From Google Earth Engine
Analysis tools	GEE and Python
Features	B2, B3, B4, B8, B11, B12, NDVI, EVI, SAVI, NDBI, elevation, slope, aspect, texture contrast, texture entropy
# points	350 Non-vegetation 350 Vegetation (150 Non-vegetation & 150 vegetation for test)

Data Preparation

- **Correlation Analysis**: Heatmap
- **Outlier removal**: Z-score
- **Feature selection:** Forward and Backward selection
- **Spatial analysis:** Grid-based, K-means, Min-distance
- **Normalization**

Correlation Analysis: Heatmap



Features removed with Correlation > 0.8:

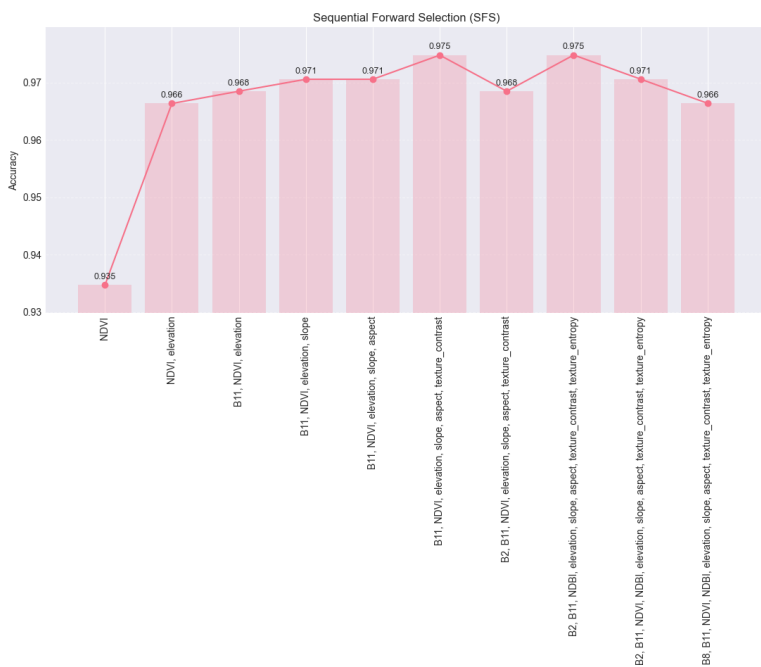
- B3
- B4
- B12
- EVI
- SAVI

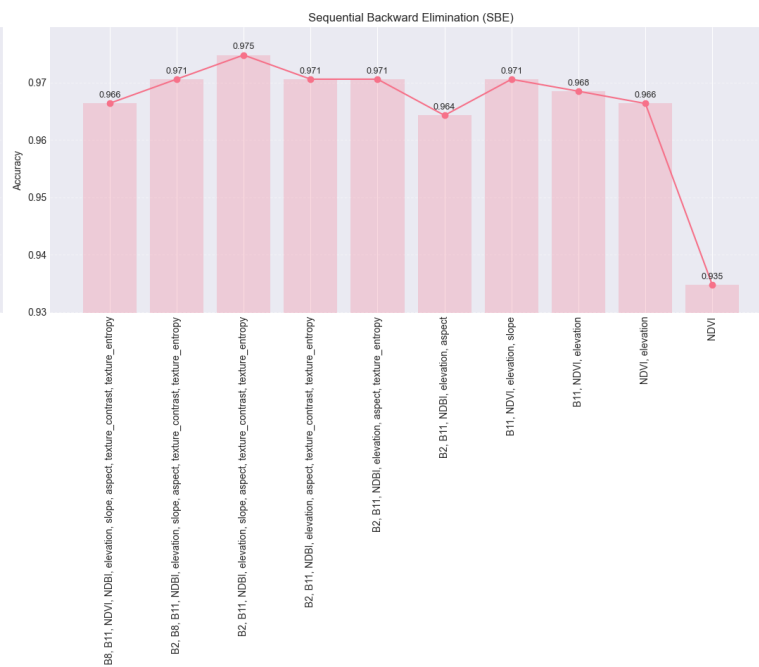
- **Outlier removal**: Z-score

Points removed with Z-score > 4	3% (21 points)
Class distribution after removal	50.8% Non-vegetation 49.2% Vegetation

- **Feature selection:** Forward and Backward selection

Random Forest best configuration	• Forward selection • B2, B11, NDBI, elevation, slope, aspect, texture contrast, texture entropy

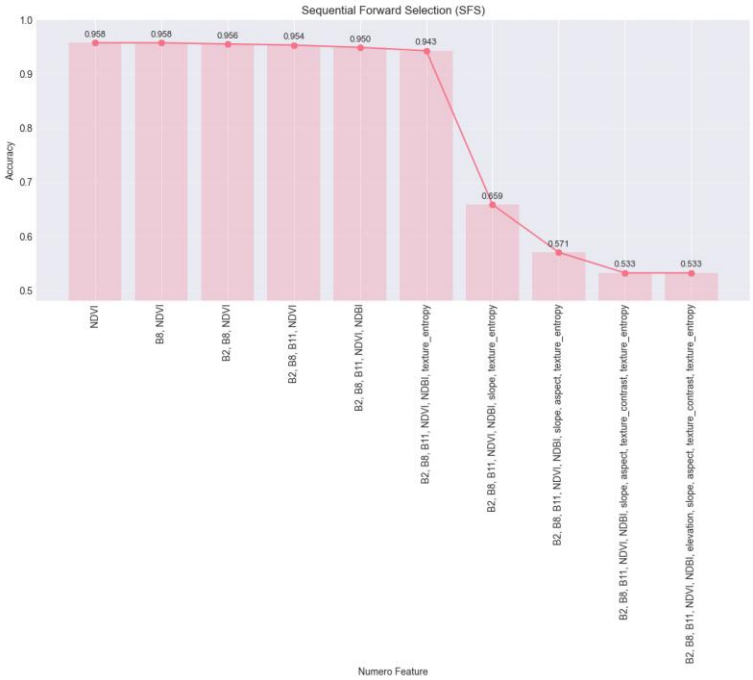


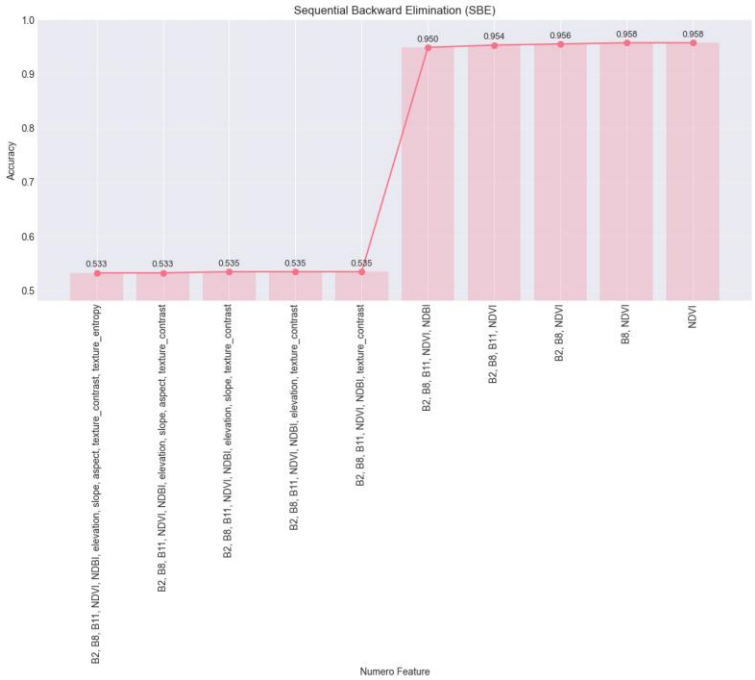


- **Feature selection:** Forward and Backward selection

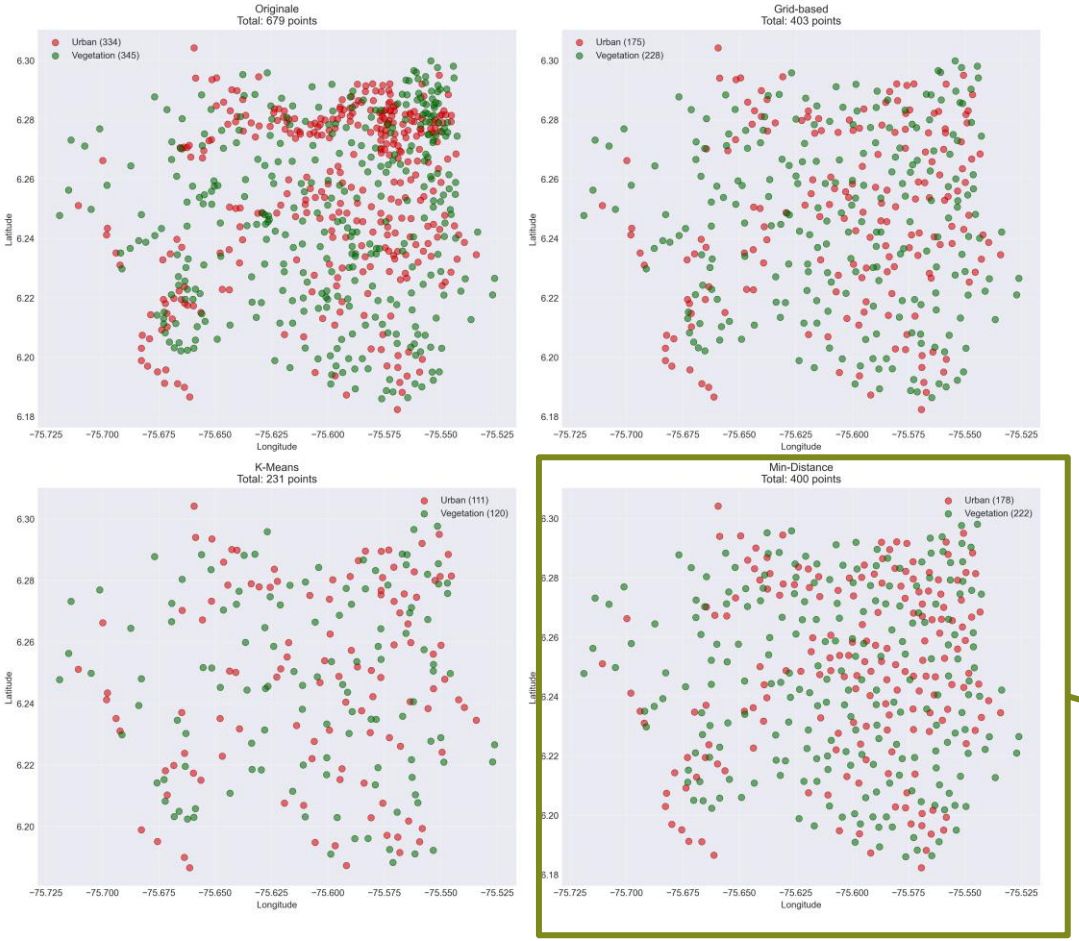
Support Vector Machine best configuration

- Backward selection
- B8, NDVI





- **Spatial analysis:** Grid-based, K-means, Min-distance



	CV
Original	0.5926
Grid-based	0.4637
K-Means	0.4768
Min-distance	0.4148

Chosen: **Min-Distance**
Non-veg. 178
Veg. 222

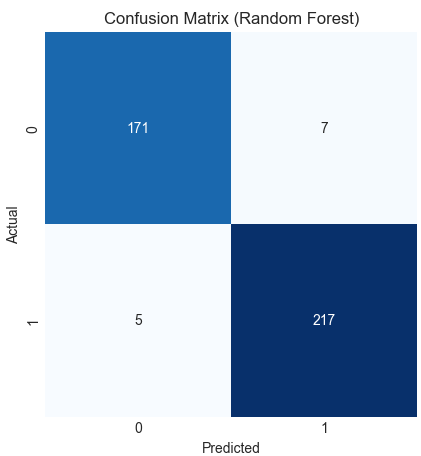
- **Normalization:** for SVM

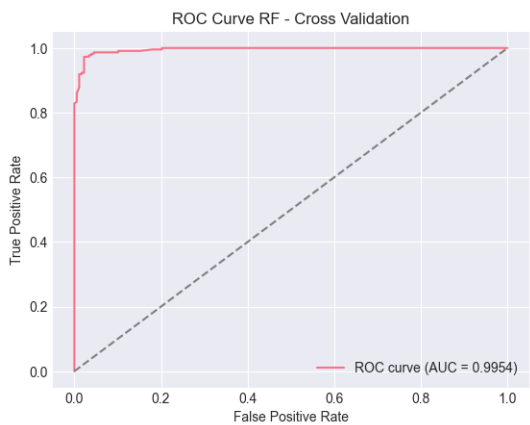
Models

	Best parameters
Random Forest	• Max depth = None • Max features = sqrt • Min sample per leaf = 1 • Min sample per split = 2 • # estimators = 100
Support Vector Machine	• C = 1 • Gamma = scale • Kernel = linear
Validation	Cross validation with 5 folds

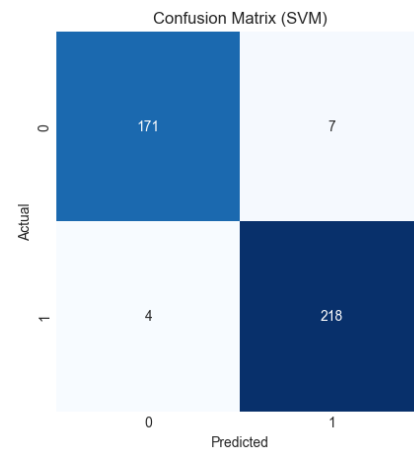
Results

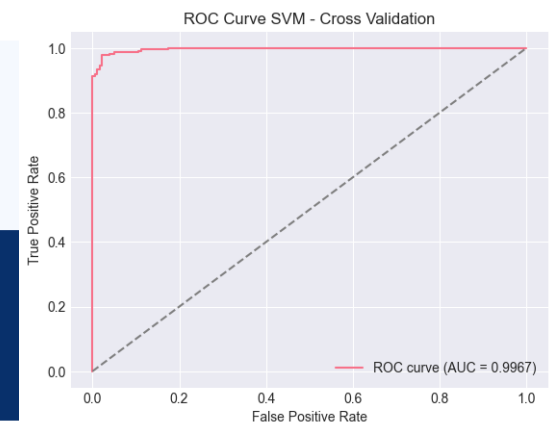
Random Forest





Support Vector Machine



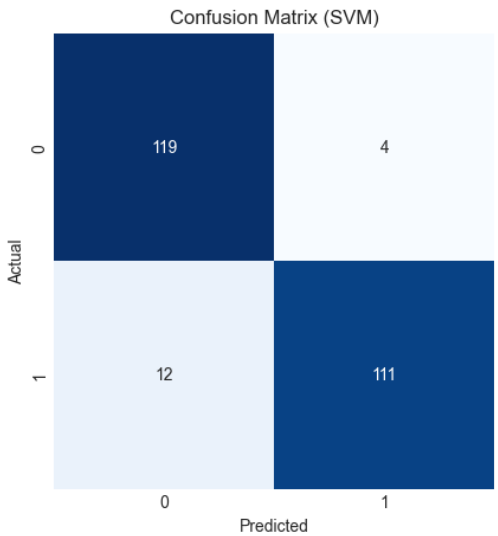


0.965	Accuracy	0.982
0.965	Precision	0.973
0.966	Recall	0.973
0.965	F1-score	0.972
0.995	AUC	0.997

Robustness Test

Test on Bogotá:

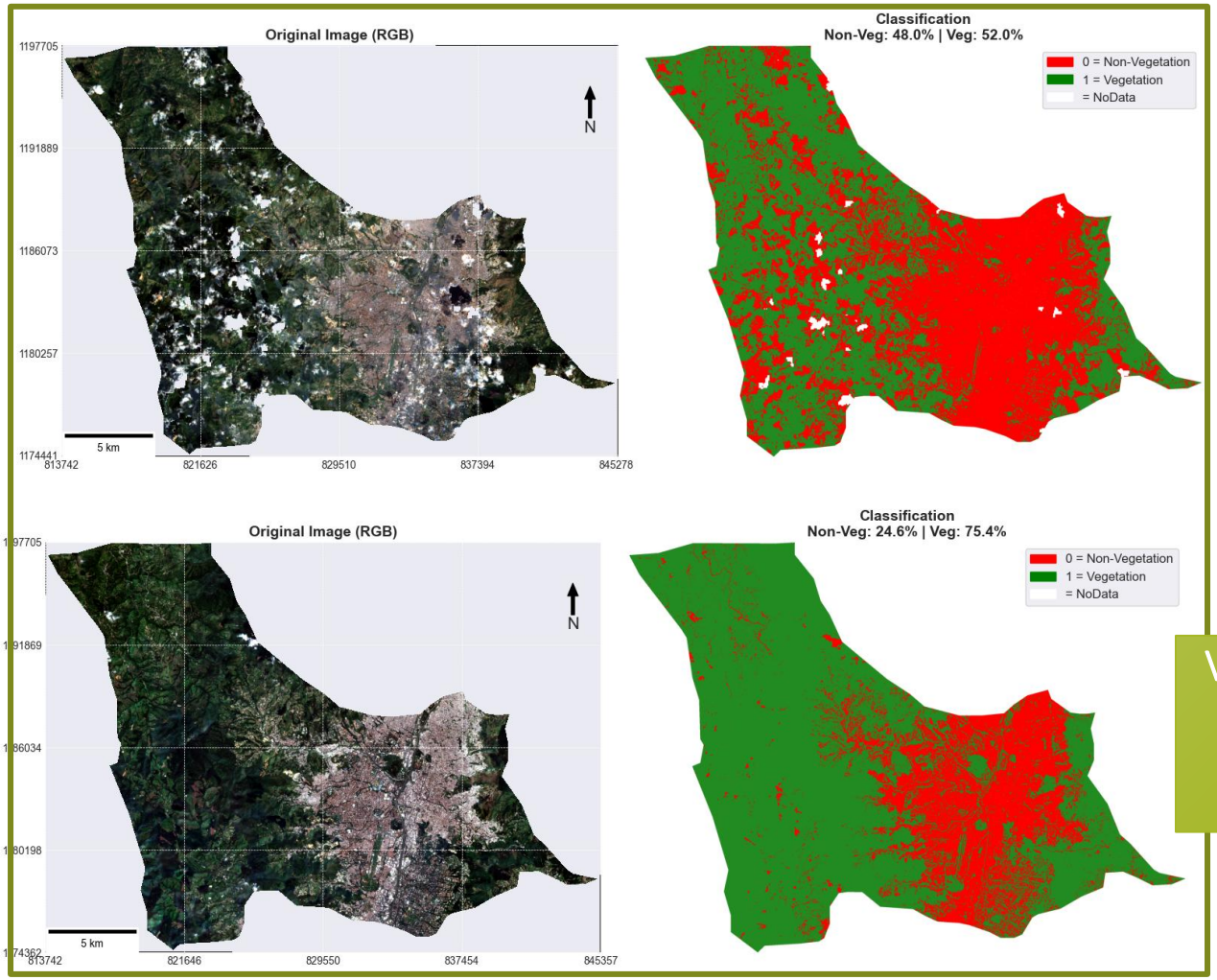
- Outlier removal with z-score method: 11 samples removed (3.67%)
- Grid-based distribution: 123 Non-Vegetation, 123 Vegetation
- Normalization
- Classification with SVM model



Results:

Accuracy	0.935
Precision	0.937
Recall	0.935
F1-score	0.935

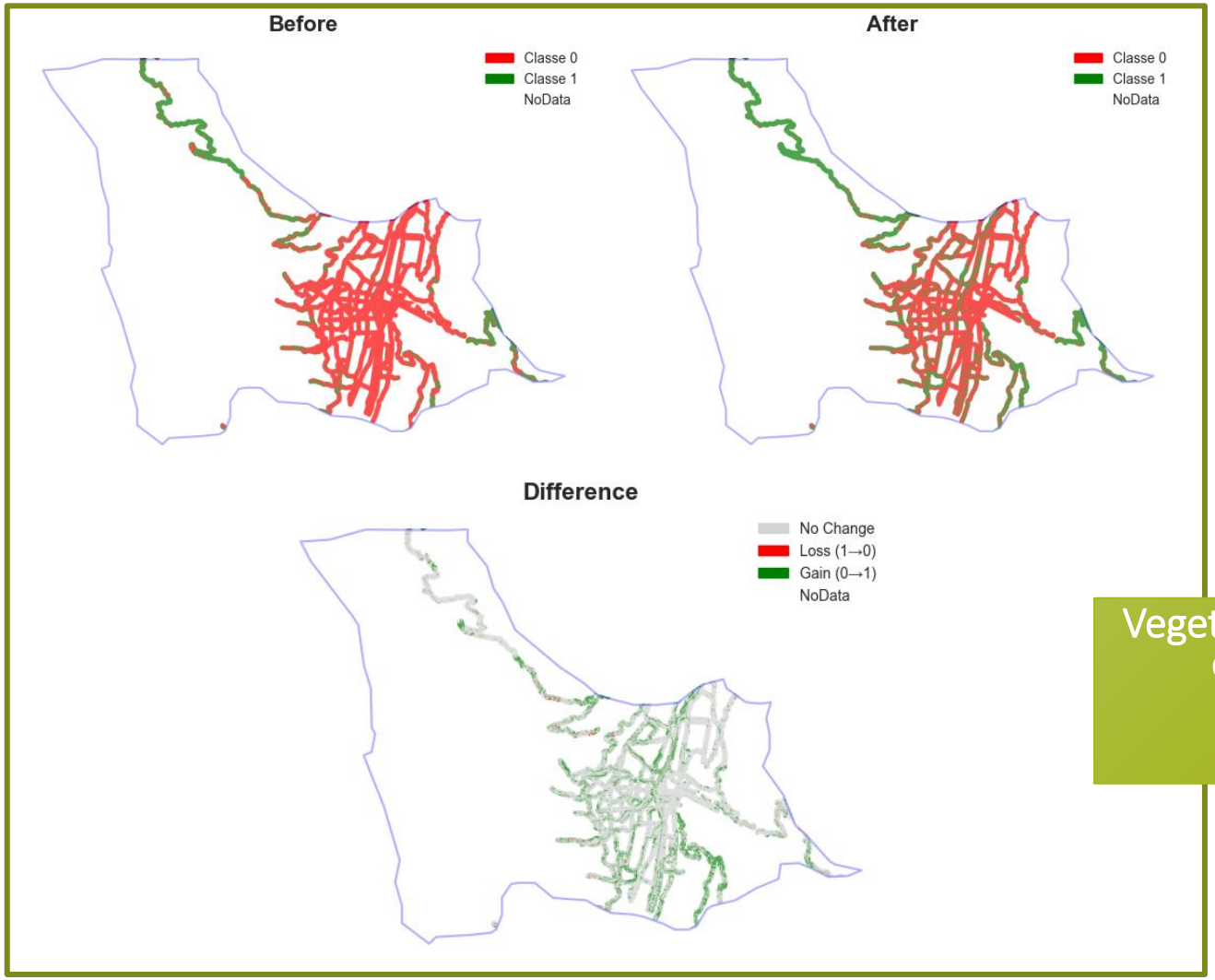
Change Detection 2016-2025



Vegetation increase in the AOI:

+46.2%

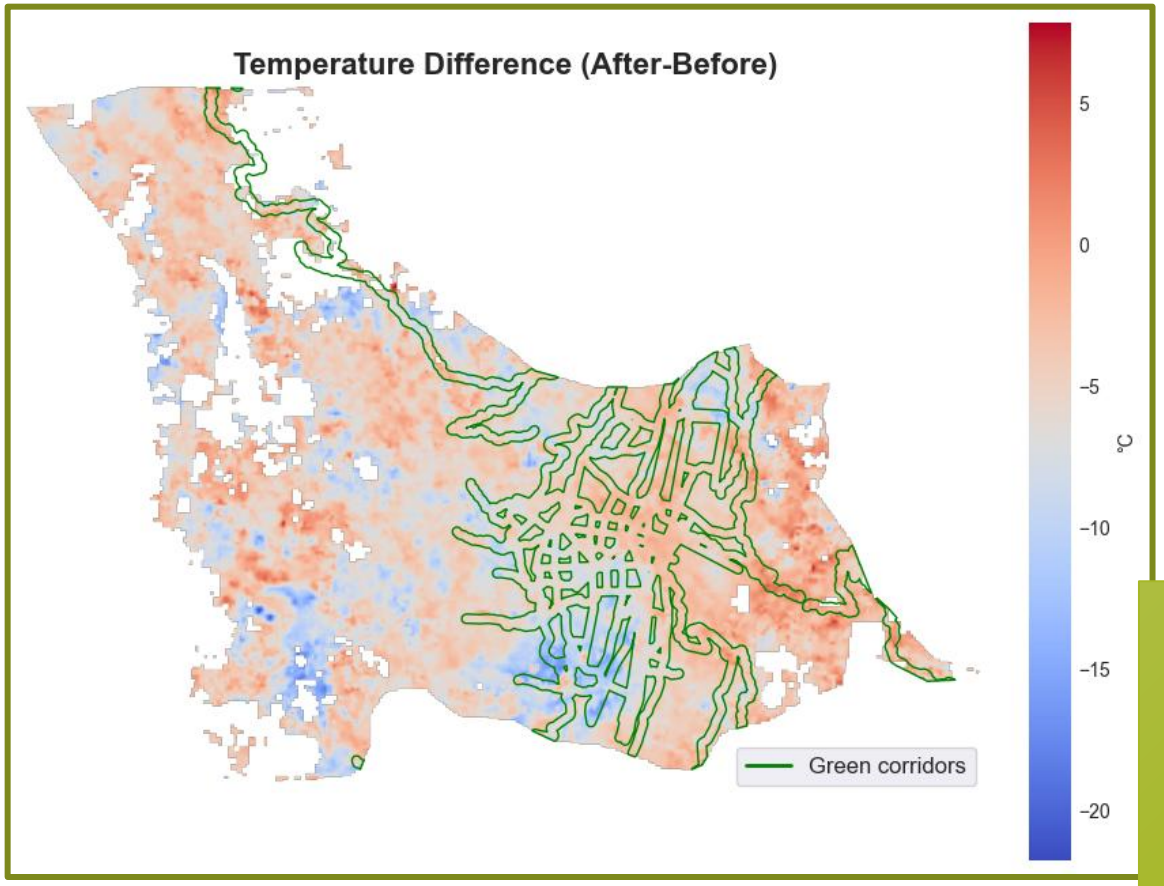
Change Detection 2016-2025



Vegetation in Green corridors:

+110.6%

Thermal Analysis



AOI average cooling:

-4.8°C

Corridors average cooling:

-5°C

Conclusions

- Urban greening is effective in urban heat island mitigation
- ML + EO robust framework for Land cover and Change detection

  both Random Forest and Support Vector Machine result suitable
- Method scalable for other subtropical cities

Limitation

- Binary classification could oversimplify the problem
- Higher resolution images could better interpret the problem
- Seasonal and interannual variability are not considered

Thanks for attention